Financial Econometrics Practice TS #2 (Forecasting)

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1 Load Packages and Data

```r
library(forecast)
library(tseries)
library(lmtest)
library(nortest)
library(MASS)
```

We read the unemployment data from the Spanish Employment Public Service (SEPE), representing the count of individuals registered as unemployed at SEPE offices. The dataset spans from January 2000 to present, sourced from https://expinterweb.mites.gob.es/series/. The data is converted into a time series object with a starting window from 2000.

```r
series <- window(
  ts(read.table("unemploy.dat", dec = ",") / 1e6,
     start = 1996, freq = 12),
  start = c(2000),
  end = c(2025, 12)
)

# Display properties
cat("Class:", class(series), "\n")
cat("Length:", length(series), "observations\n")
cat("Start:", start(series)[1], "Period", start(series)[2], "\n")
```

```r
cat("End:", end(series)[1], "Period", end(series)[2], "\n")
cat("Frequency:", frequency(series), "\n\n")

# Display first five years of data
head(round(series, 3),12*5)
```

```
## Class: ts
## Length: 312 observations
## Start: 2000 Period 1
## End: 2025 Period 12
## Frequency: 12
##
##        Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
## 2000 2.090 2.080 2.042 1.984 1.929 1.903 1.893 1.888 1.908 1.934 1.963 1.947
## 2001 2.017 1.993 1.981 1.910 1.898 1.843 1.836 1.879 1.889 1.941 1.986 1.989
## 2002 2.075 2.150 2.083 2.060 2.003 1.963 1.962 1.984 2.007 2.065 2.117 2.127
## 2003 2.185 2.180 2.163 2.104 2.036 2.020 1.996 2.017 2.040 2.097 2.143 2.181
## 2004 2.233 2.219 2.182 2.162 2.091 2.054 2.014 2.050 2.051 2.076 2.121 2.113
```

```r
# constants
series_name <- "Unemployment Rate"
start_year <- 2000
start_period <- 1
freq <- 12
period_length <- 12
xlabel <- "month"
ylabel <- "Unemployment Rate (%)"
```

1.1 Plotting the series

```r
plot(series, main=series_name, xlab=xlabel, ylab=ylabel)
abline(v=1996:2026,col=4,lty=3)
```

The data exhibits clear yearly seasonal patterns with peaks at end of the year and troughs in the middle of the year, likely due to seasonal employment patterns. We also see changes in trend from 2000 to 2008, then a sharp increase during the 2008 financial crisis, followed by a gradual decline until 2020, and then a sharp increase again during the COVID-19 pandemic.

To fit a SARIMA model, we will transform the series to achieve mean and variance stability. We begin with a preliminary visual evaluation of stationarity by analyzing rolling mean and variance. This approach provides a continuous view of potential changes, avoiding the artifacts introduced by binning.

2 Transformations

2.1 Functions

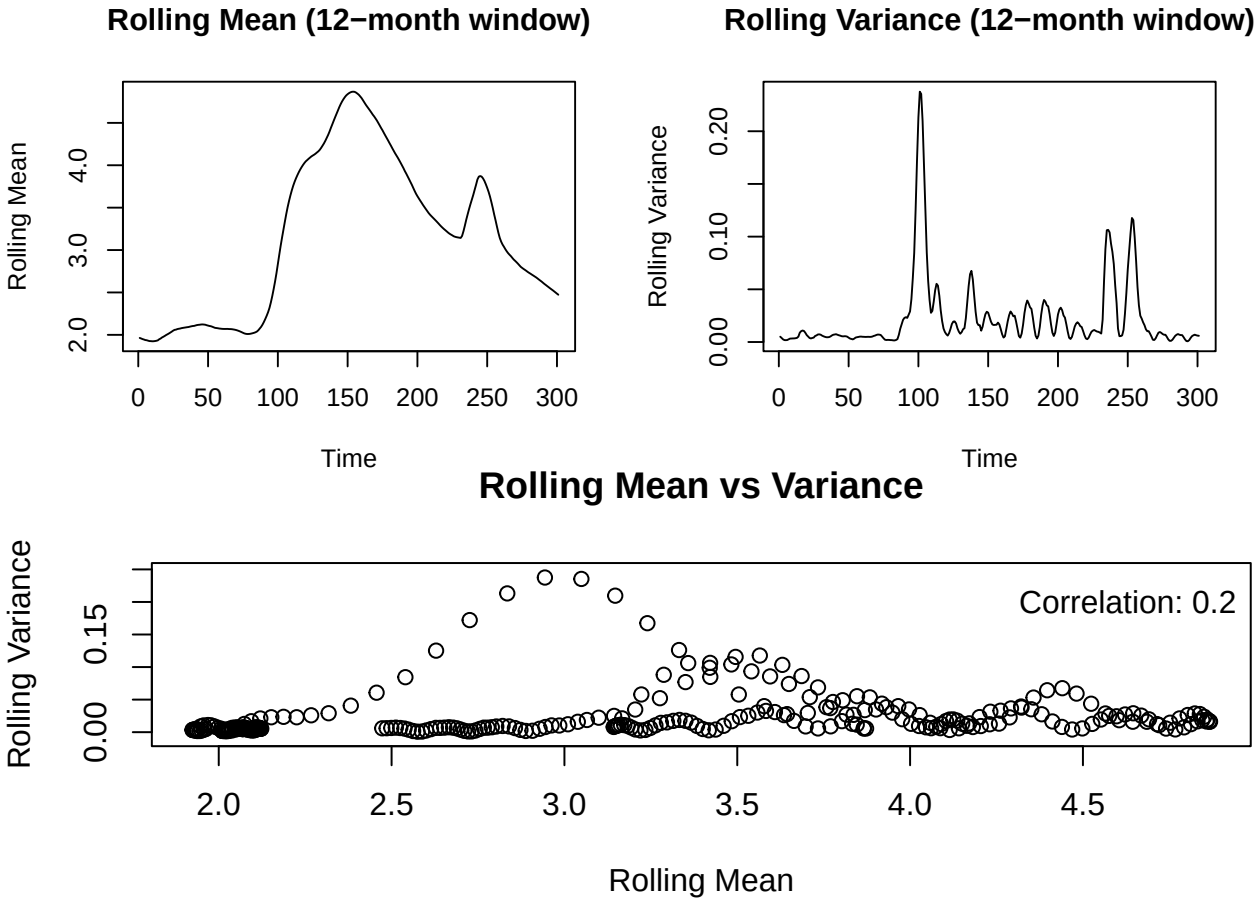
```r
# function to get rolling ts
get_rolling_ts <- function(series, window_size) {
  rolling_mean <- sapply(1:(length(series) - window_size + 1), function(i) {
    mean(series[i:(i + window_size - 1)])
  })
  rolling_var <- sapply(1:(length(series) - window_size + 1), function(i) {
    var(series[i:(i + window_size - 1)])
  })

  data.frame(
    rolling_mean = rolling_mean,
    rolling_variance = rolling_var
  )
}

produce_plots <- function(series, name="Series") {
  rolling_stats <- get_rolling_ts(series, freq)
  print(paste("Rolling statistics for:", name))
  par(mfrow=c(2,2))

  # Plot 1: Rolling Mean over time
  plot(rolling_stats$rolling_mean, type="l",
       main = paste("Rolling Mean (", freq, "-month window)", sep = ""),
       xlab = "Time", ylab = "Rolling Mean")

  # Plot 2: Rolling Variance over time
  plot(rolling_stats$rolling_variance, type="l",
       main = paste("Rolling Variance (", freq, "-month window)", sep = ""),
       xlab = "Time", ylab = "Rolling Variance")

  # Plot 3 & 4: Mean vs Variance (spanning full width)
  par(mfrow=c(2,1), oma=c(0, 0, 2, 0))
  y_range <- range(rolling_stats$rolling_variance)
  y_padding <- (y_range[2] - y_range[1]) * 0.05
  plot(rolling_stats$rolling_mean, rolling_stats$rolling_variance,
       main = "Rolling Mean vs Variance",
       xlab = "Rolling Mean", ylab = "Rolling Variance",
       ylim = c(y_range[1] - y_padding, y_range[2] + y_padding))

  # print correlation
  cor_val <- cor(rolling_stats$rolling_mean, rolling_stats$rolling_variance)
```

```
  legend("topright", legend = paste("Correlation:", round(cor_val, 3)), bty = "n")

  par(mfrow=c(1,1))
}
```

2.2 Variance and Mean Stability Check

```
produce_plots (series, series_name)
```

```
## [1] "Rolling statistics for: Unemployment Rate"
```



We can see that the rolling mean is not stable, with clear changes in trend over time. The rolling variance also shows significant fluctuations, particularly during the 2008 financial crisis and the COVID-19 pandemic. The correlation between rolling mean and variance is low (0.08), suggesting that the variance does not strongly depend on the mean, but the presence of heteroskedasticity is still evident from the variance plot.

We will first address the variance instability with transformations, and then we will address the mean instability with differencing prior to fitting the SARIMA model.

2.3 LN and Box-Cox Transformations

```
ln_transformed <- log(series / 100)

# Box-Cox transformation
boxcox_result <- boxcox(series ~ 1, lambda = seq(-2, 2, by = 0.01), plot=FALSE)
```

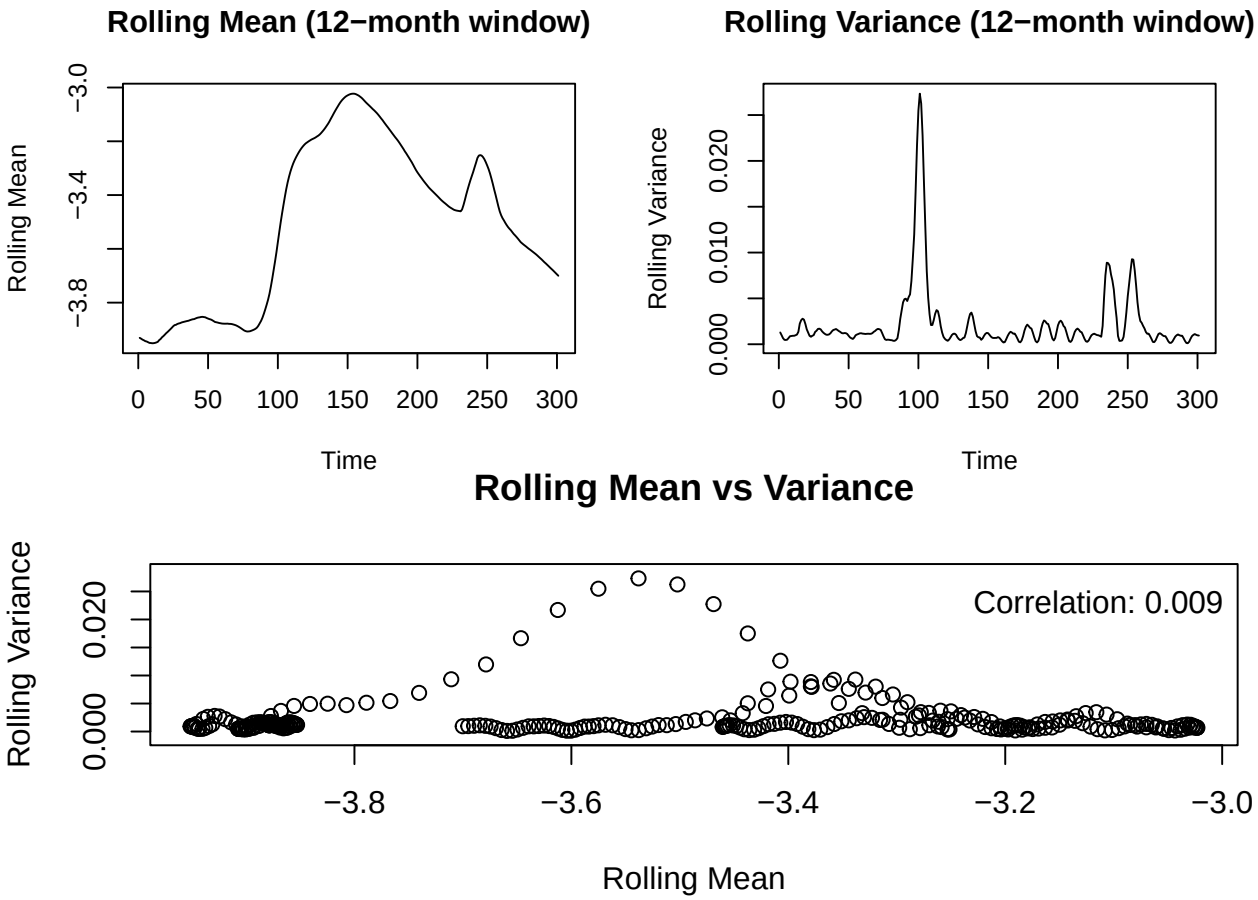
```r
optimal_lambda <- boxcox_result$x[which.max(boxcox_result$y)]
cat("Optimal lambda for Box-Cox transformation:", optimal_lambda, "\n")
# Apply Box-Cox transformation with optimal lambda
bc_transformed <- (series^optimal_lambda - 1) / optimal_lambda
```

```
## Optimal lambda for Box-Cox transformation: -0.03
```

Because optimal lambda for the Box-Cox transformation is close to 0, the log transformation is likely sufficient for variance stabilization. Thus, we will proceed to investigate the plots only for the log transformation.

```r
produce_plots (ln_transformed, paste(series_name, "(log transformed)"))
```

```
## [1] "Rolling statistics for: Unemployment Rate (log transformed)"
```



Transforming the data with the natural logarithm reduced the overall level butdid not stabilize the variance. The 2008 financial crisis and the COVID-19 pandemic still show significant spikes in variance, which suggest that there could be structural breaks in the data.

For this exercise, we keep as is as handling structural breaks is beyond the scope of this assignment.

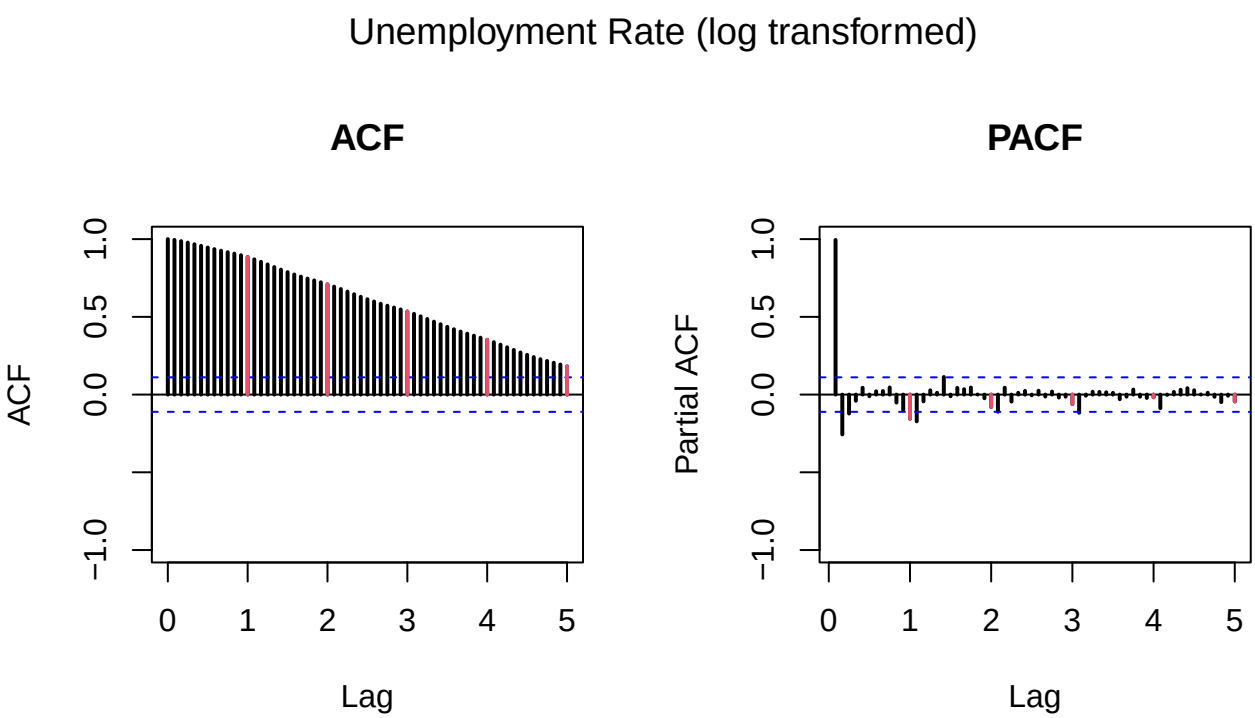
2.4 ACF and PACF Analysis

Using the log-transformed series, we will analyze the ACF and PACF plots to determine the appropriate order of differencing and the potential AR and MA terms for the SARIMA model.

```r
acf_pacf_plots <- function(series, name="Series") {
  s <- frequency(series)
  lag_limit <- max(60, 3*s)
```

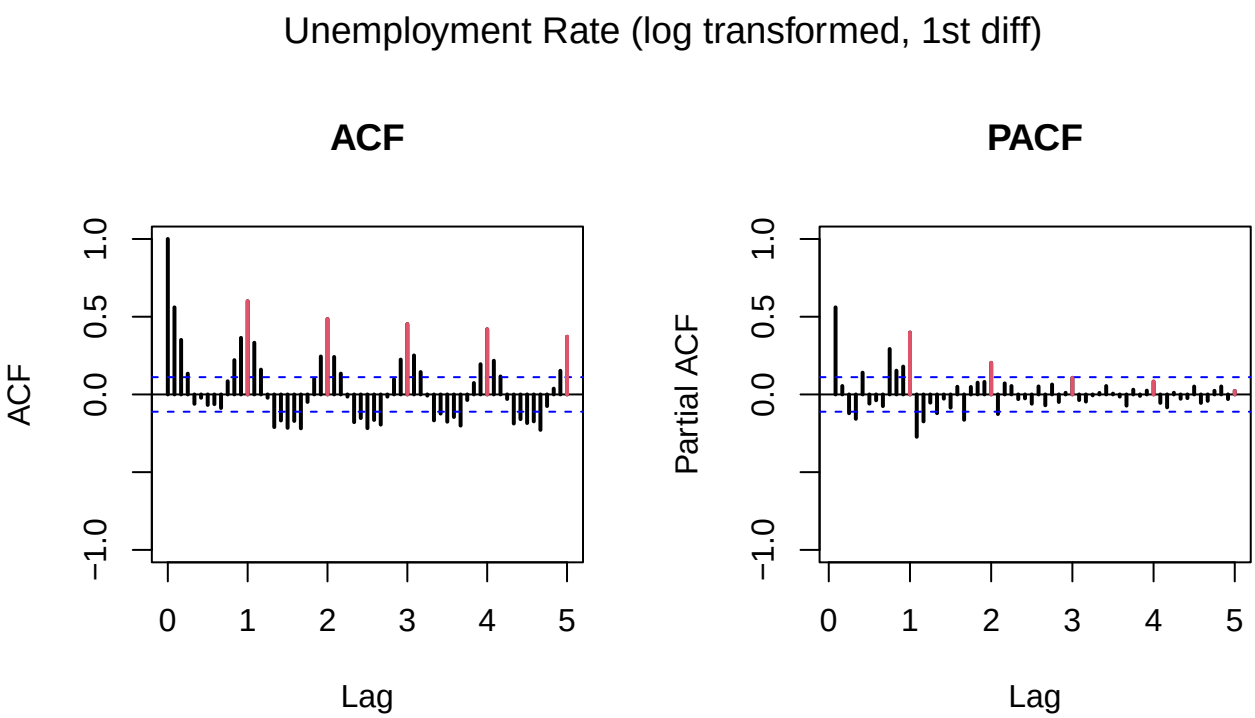
```r
  acf_obj <- acf(series, plot=FALSE, lag.max=lag_limit)
  pacf_obj <- pacf(series, plot=FALSE, lag.max=lag_limit)

  par(mfrow=c(1,2), oma=c(0, 0, 2, 0))
  plot(acf_obj, main = "ACF", ylim=c(-1,1), lwd=2, col=1)

  acf_lags <- as.numeric(acf_obj$lag)[-1]
  acf_vals <- as.numeric(acf_obj$acf)[-1]
  seasonal_lags <- seq(s, lag_limit, by=s)

  acf_idx <- which(round(acf_lags * s) %in% seasonal_lags)
  segments(acf_lags[acf_idx], 0, acf_lags[acf_idx], acf_vals[acf_idx], col=2, lwd=2)
  plot(pacf_obj, main = "PACF", ylim=c(-1,1), lwd=2, col=1)

  pacf_lags <- as.numeric(pacf_obj$lag)
  pacf_vals <- as.numeric(pacf_obj$acf)
  pacf_idx <- which(round(pacf_lags * s) %in% seasonal_lags)
  segments(pacf_lags[pacf_idx], 0, pacf_lags[pacf_idx], pacf_vals[pacf_idx], col=2, lwd=2)

  mtext(name, outer=TRUE, cex=1.2, line=0.5)
  par(mfrow = c(1, 1))
}
```

```r
acf_pacf_plots(ln_transformed, paste(series_name, "(log transformed)"))
```



The ACF plot shows slow decay with all lag significant which suggests non-stationarity and the need for differencing. We apply first differencing and then re-examine the ACF and PACF plots.

```r
d1ln_transformed <- diff(ln_transformed)
acf_pacf_plots(d1ln_transformed, paste(series_name, "(log transformed, 1st diff)"))
```



The PACF plot has significant nearby lags at 3, 4, 5, and seasonal lag 12. The ACF plot has significant nearby lags at 1, 2, and seasonal lag 12. From here, we will consider models with AR terms up to 5, MA terms up to 3, and seasonal AR and MA terms at lag 12.

3 SARIMA Models

3.1 Functions

```
analyze_sarima <- function(series, order, seasonal, verbose = TRUE) {

  s <- frequency(series)
  n <- length(series)
  model_name <- paste0("ARIMA(", paste(order, collapse=","), ")(",
                       paste(seasonal$order, collapse=","), ")", s)

  if (verbose) {
    cat("\n==================================================================\n")
    cat("  ANALYSIS FOR MODEL:", model_name, "\n")
    cat("==================================================================\n")
  }

  # 1. Estimation & Significance Check
  fit_full <- arima(series, order = order, seasonal = seasonal)

  diag_var <- diag(fit_full$var.coef)
  if (any(diag_var <= 0)) {
    if (verbose) {
      cat("\n   CRITICAL WARNING: The Hessian matrix is unstable (Negative Variances).\n")
    }
    se_vec <- ifelse(diag_var > 0, sqrt(diag_var), NA)
  } else {
```

```r
  se_vec <- sqrt(diag_var)
}

t_vec  <- fit_full$coef / se_vec
p_vec  <- 2 * (1 - pnorm(abs(t_vec)))

coeftest <- data.frame(
  Estimate = fit_full$coef,
  Std.Error = se_vec,
  t_stat = t_vec,
  p_value = p_vec
)

coeftest$Significant <- ifelse(
  !is.na(coeftest$p_value) &
    (coeftest$p_value < 0.05 | abs(coeftest$t_stat) > 2),
  "Yes", "No"
)

if (verbose) {
  cat("\n[1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):\n")
  print(round(coeftest[,1:4], 4))
  cat("\nSignificance Flags:\n")
  print(coeftest[,5, drop=FALSE])
}

# 2. Residual Analysis (Diagnostics)
resid <- residuals(fit_full)

# A. Causality / Invertibility Roots (Min Modulus)
ar_poly <- fit_full$model$phi
if (length(ar_poly) > 0 && any(ar_poly != 0)) {
  ar_roots_mod <- Mod(polyroot(c(1, -ar_poly)))
  min_ar_root  <- min(ar_roots_mod)
} else {
  min_ar_root  <- NA
}
ma_poly <- fit_full$model$theta
if (length(ma_poly) > 0 && any(ma_poly != 0)) {
  ma_roots_mod <- Mod(polyroot(c(1, ma_poly)))
  min_ma_root  <- min(ma_roots_mod)
} else {
  min_ma_root  <- NA
}

is_causal     <- ifelse(!is.na(min_ar_root), min_ar_root > 1, TRUE)
is_invertible <- ifelse(!is.na(min_ma_root), min_ma_root > 1, TRUE)

if (verbose) {
  cat("\n[2] DIAGNOSTICS:\n")
  cat("Min AR Root Modulus: ", min_ar_root, "\n")
  cat("Min MA Root Modulus: ", min_ma_root, "\n")
}
```

```r
  # B. Plots
  if (verbose) {
    par(mfrow = c(2, 2), mar = c(4, 4, 3, 2))
    plot(resid, main = "Residuals", ylab = "Error")
    abline(h = 0, col = "gray")
    abline(h = c(-3 * sd(resid), 3 * sd(resid)), lty = 2, col = "red")
    scatter.smooth(sqrt(abs(resid)), main = "Variance Check",
                   ylab = "sqrt(|Resid|)", lpars = list(col = "red"))
    qqnorm(resid)
    qqline(resid, col = "red")
    hist(resid, breaks = 20, freq = FALSE, main = "Histogram")
    curve(dnorm(x, mean = mean(resid), sd = sd(resid)),
          col = "red", add = TRUE)
  }

  # C. Tests
  sw_test <- shapiro.test(resid)
  lb_test <- Box.test(resid, lag = 20, type = "Ljung-Box")

  # 3. Forecasting & Cross-Validation
  train_end <- c(floor(time(series)[n-12]), cycle(series)[n-12])
  train_set <- window(series, end = train_end)
  test_set  <- window(series, start = time(series)[n-11])

  fit_train <- arima(train_set, order = order, seasonal = seasonal)
  fc <- predict(fit_train, n.ahead = 12)

  fc_orig <- exp(fc$pred); test_orig <- exp(test_set)
  lower_bound <- exp(fc$pred - 1.96 * fc$se); upper_bound <- exp(fc$pred + 1.96 * fc$se)

  # 4. Summary Table Generation
  summary_row <- data.frame(
    Model_Name      = model_name,
    AIC             = fit_full$aic,
    BIC             = BIC(fit_full),
    Sigma2          = fit_full$sigma2,
    Resid_Norm_PVal = sw_test$p.value,
    Normal       = ifelse(sw_test$p.value > 0.05, "Yes", "No"),
    Resid_Ind_PVal  = lb_test$p.value,
    Independent  = ifelse(lb_test$p.value > 0.05, "Yes", "No"),
    Min_AR_Root     = min_ar_root,
    Causal          = ifelse(is_causal, "Yes", "No"),
    Min_MA_Root     = min_ma_root,
    Invertible      = ifelse(is_invertible, "Yes", "No"),
    RMSE            = sqrt(mean((test_orig - fc_orig)^2)),
    RMSPE           = sqrt(mean(((test_orig - fc_orig) / test_orig)^2)) * 100,
    MAE             = mean(abs(test_orig - fc_orig)),
    MAPE_Pct        = mean(abs((test_orig - fc_orig) / test_orig)) * 100,
    Mean_CI_Width   = mean(upper_bound - lower_bound)
  )
  return(summary_row)
}
```

```r
compare_sarima_models <- function(series, models_list, verbose = TRUE) {
  if (verbose) {
    cat("\n==================================================================\n")
    cat("  STARTING AUTOMATED MODEL COMPARISON\n")
    cat("==================================================================\n")
  }

  comparison_results <- list()
  for (i in seq_along(models_list)) {
    spec <- models_list[[i]]
    if (verbose) {
      cat(paste0("\n==================================================================\n"))
      cat(paste0("  MODEL ", i, ": ARIMA(", paste(spec$order, collapse=","), ")(",
                 paste(spec$seasonal$order, collapse=","), ")", frequency(series), "\n"))
      cat(paste0("==================================================================\n"))
    }
    result <- tryCatch({
      analyze_sarima(series = series,
                     order = spec$order,
                     seasonal = spec$seasonal,
                     verbose = verbose)
    }, error = function(e) {
      cat(paste0("  Error in Model ", i, ": ", e$message, "\n"))
      return(NULL) # Return NULL so we can skip it
    })
    if (!is.null(result)) {
      comparison_results[[length(comparison_results) + 1]] <- result
    }
  }

  final_table <- do.call(rbind, comparison_results)

  cat("\n--- Comparison Complete ---\n")
  return(final_table)
}
```

3.2 Models

We list all possibl models with non-seasonal AR terms up to 5, non-seasonal MA terms up to 3, and seasonal AR and MA terms at lag 12. This results in a total of 24 models to compare.

```r
# All possible non-seasonal ARMA(p,q) with p <= 5 and q <= 3
# Seasonal period fixed at 12
model_grid <- expand.grid(p = 0:5, q = 0:3)

# Create all model names first
model_names <- paste0(
  "ARIMA(", model_grid$p, ",0,", model_grid$q, ")",
  "(1,0,1)[12]"
)

# Create my_models following the same order with names
my_models <- setNames(
  lapply(seq_len(nrow(model_grid)), function(i) {
```

```r
    list(
      order = c(model_grid$p[i], 0, model_grid$q[i]),
      seasonal = list(order = c(1,0,1), period = 12)
    )
  }),
  model_names
)

print(model_names)
```

```
##  [1] "ARIMA(0,0,0)(1,0,1)[12]" "ARIMA(1,0,0)(1,0,1)[12]"
##  [3] "ARIMA(2,0,0)(1,0,1)[12]" "ARIMA(3,0,0)(1,0,1)[12]"
##  [5] "ARIMA(4,0,0)(1,0,1)[12]" "ARIMA(5,0,0)(1,0,1)[12]"
##  [7] "ARIMA(0,0,1)(1,0,1)[12]" "ARIMA(1,0,1)(1,0,1)[12]"
##  [9] "ARIMA(2,0,1)(1,0,1)[12]" "ARIMA(3,0,1)(1,0,1)[12]"
## [11] "ARIMA(4,0,1)(1,0,1)[12]" "ARIMA(5,0,1)(1,0,1)[12]"
## [13] "ARIMA(0,0,2)(1,0,1)[12]" "ARIMA(1,0,2)(1,0,1)[12]"
## [15] "ARIMA(2,0,2)(1,0,1)[12]" "ARIMA(3,0,2)(1,0,1)[12]"
## [17] "ARIMA(4,0,2)(1,0,1)[12]" "ARIMA(5,0,2)(1,0,1)[12]"
## [19] "ARIMA(0,0,3)(1,0,1)[12]" "ARIMA(1,0,3)(1,0,1)[12]"
## [21] "ARIMA(2,0,3)(1,0,1)[12]" "ARIMA(3,0,3)(1,0,1)[12]"
## [23] "ARIMA(4,0,3)(1,0,1)[12]" "ARIMA(5,0,3)(1,0,1)[12]"
```

3.3 Run Analysis

```r
# Run Analysis
comparison_table <- compare_sarima_models(
  d1ln_transformed,
  my_models,
  verbose = FALSE
)
print(comparison_table)
```

```
##   Error in Model 21: non-finite finite-difference value [2]
##
## --- Comparison Complete ---
##                Model_Name       AIC       BIC       Sigma2 Resid_Norm_PVal
## 1  ARIMA(0,0,0)(1,0,1)12 -1689.553 -1674.593 0.0002399648    3.118229e-16
## 2  ARIMA(1,0,0)(1,0,1)12 -1830.368 -1811.669 0.0001476447    3.367757e-14
## 3  ARIMA(2,0,0)(1,0,1)12 -1833.604 -1811.165 0.0001453623    1.192058e-14
## 4  ARIMA(3,0,0)(1,0,1)12 -1831.660 -1805.482 0.0001453355    1.174299e-14
## 5  ARIMA(4,0,0)(1,0,1)12 -1833.278 -1803.360 0.0001429728    6.308674e-15
## 6  ARIMA(5,0,0)(1,0,1)12 -1836.572 -1802.914 0.0001403430    9.728857e-15
## 7  ARIMA(0,0,1)(1,0,1)12 -1776.737 -1758.038 0.0001780468    1.513239e-13
## 8  ARIMA(1,0,1)(1,0,1)12 -1833.625 -1811.186 0.0001452986    1.170265e-14
## 9  ARIMA(2,0,1)(1,0,1)12 -1835.071 -1808.892 0.0001429969    9.511915e-15
## 10 ARIMA(3,0,1)(1,0,1)12 -1830.253 -1800.334 0.0001448423    9.049270e-15
## 11 ARIMA(4,0,1)(1,0,1)12 -1836.007 -1802.349 0.0001404706    6.731279e-15
## 12 ARIMA(5,0,1)(1,0,1)12 -1834.788 -1797.390 0.0001402015    9.262329e-15
## 13 ARIMA(0,0,2)(1,0,1)12 -1817.293 -1794.854 0.0001536828    1.404421e-14
## 14 ARIMA(1,0,2)(1,0,1)12 -1832.079 -1805.901 0.0001447888    9.770616e-15
## 15 ARIMA(2,0,2)(1,0,1)12 -1833.228 -1803.310 0.0001430103    8.844384e-15
## 16 ARIMA(3,0,2)(1,0,1)12 -1833.005 -1799.347 0.0001420242    4.884623e-15
## 17 ARIMA(4,0,2)(1,0,1)12 -1832.958 -1795.560 0.0001415366    1.284736e-14
```

```
## 18 ARIMA(5,0,2)(1,0,1)12 -1833.762 -1792.624 0.0001402090    2.311044e-14
## 19 ARIMA(0,0,3)(1,0,1)12 -1825.312 -1799.133 0.0001487181    2.595307e-14
## 20 ARIMA(1,0,3)(1,0,1)12 -1837.381 -1807.463 0.0001408264    5.371190e-15
## 21 ARIMA(3,0,3)(1,0,1)12 -1837.647 -1800.249 0.0001388844    1.043977e-14
## 22 ARIMA(4,0,3)(1,0,1)12 -1830.114 -1788.977 0.0001412112    1.581858e-14
## 23 ARIMA(5,0,3)(1,0,1)12 -1835.399 -1790.521 0.0001380309    1.257154e-14
##    Normal Resid_Ind_PVal Independent Min_AR_Root Causal Min_MA_Root Invertible
## 1      No   0.000000e+00          No    1.002981    Yes    1.026800        Yes
## 2      No   9.928266e-02         Yes    1.000640    Yes    1.012133        Yes
## 3      No   3.322451e-01         Yes    1.000723    Yes    1.013171        Yes
## 4      No   3.334708e-01         Yes    1.000719    Yes    1.013101        Yes
## 5      No   4.624282e-01         Yes    1.000609    Yes    1.012648        Yes
## 6      No   7.358013e-01         Yes    1.000566    Yes    1.012156        Yes
## 7      No   0.000000e+00          No    1.001365    Yes    1.017180        Yes
## 8      No   3.330666e-01         Yes    1.000721    Yes    1.013275        Yes
## 9      No   4.642005e-01         Yes    1.000582    Yes    1.012163        Yes
## 10     No   3.618883e-01         Yes    1.000679    Yes    1.012908        Yes
## 11     No   6.972193e-01         Yes    1.000541    Yes    1.011875        Yes
## 12     No   7.424442e-01         Yes    1.000561    Yes    1.012133        Yes
## 13     No   2.529262e-06          No    1.000820    Yes    1.013720        Yes
## 14     No   3.321379e-01         Yes    1.000682    Yes    1.013272        Yes
## 15     No   4.780029e-01         Yes    1.000597    Yes    1.012262        Yes
## 16     No   5.779688e-01         Yes    1.000574    Yes    1.012173        Yes
## 17     No   6.870367e-01         Yes    1.000619    Yes    1.011946        Yes
## 18     No   8.359843e-01         Yes    1.000641    Yes    1.012633        Yes
## 19     No   1.592004e-02          No    1.000742    Yes    1.012547        Yes
## 20     No   6.831260e-01         Yes    1.000541    Yes    1.011715        Yes
## 21     No   8.788881e-01         Yes    1.000567    Yes    1.012216        Yes
## 22     No   5.736730e-01         Yes    1.000594    Yes    1.008526        Yes
## 23     No   9.222191e-01         Yes    1.000526    Yes    1.011350        Yes
##           RMSE     RMSPE         MAE  MAPE_Pct Mean_CI_Width
## 1  0.005155738 0.5231994 0.003961321 0.3999178    0.06157781
## 2  0.006160290 0.6223465 0.005002779 0.5034827    0.05957034
## 3  0.005933486 0.5997874 0.004760245 0.4791940    0.05890885
## 4  0.005937267 0.6002102 0.004767790 0.4799744    0.05897198
## 5  0.005940510 0.6000381 0.004725319 0.4754416    0.05840570
## 6  0.005868519 0.5922004 0.004690942 0.4717455    0.05756988
## 7  0.005787495 0.5857309 0.004633812 0.4668552    0.05751429
## 8  0.005897894 0.5961513 0.004715277 0.4746397    0.05873444
## 9  0.005911938 0.5968776 0.004688363 0.4715944    0.05737218
## 10 0.005984126 0.6048544 0.004806914 0.4838689    0.05896610
## 11 0.006001468 0.6056805 0.004795174 0.4822412    0.05701489
## 12 0.005917737 0.5970759 0.004740685 0.4767073    0.05733844
## 13 0.006051573 0.6119102 0.004926986 0.4960962    0.05693339
## 14 0.005861554 0.5922128 0.004647917 0.4677090    0.05804644
## 15 0.005894541 0.5951701 0.004670329 0.4698026    0.05741869
## 16 0.005922708 0.5980141 0.004684832 0.4712502    0.05740074
## 17 0.006196291 0.6263256 0.004882362 0.4911521    0.05867902
## 18 0.005867103 0.5923457 0.004720548 0.4749116    0.05830403
## 19 0.006185493 0.6251894 0.005069440 0.5103412    0.05872291
## 20 0.005929427 0.5985817 0.004714886 0.4742491    0.05732075
## 21 0.005873528 0.5927018 0.004686809 0.4713145    0.05711622
## 22 0.006029664 0.6088515 0.004729350 0.4755219    0.05722668
## 23 0.006045785 0.6096975 0.004806207 0.4832333    0.05703823
```

3.4 Choosing Model

From the comparison table we can see that:

- All models are causal and invertible
- None of the models have normal residuals, likely due to the structural breaks in the data caused by the 2008 financial crisis and the COVID-19 pandemic.
- Most models have independent residuals, with a few exceptions.
- We can see that the performance metrics produce generally similar ranking across models.

We select the top two models based on BIC, which provides a stronger penalty for model complexity than AIC, and on RMSPE, which is preferred over MAPE_pct due to its standardized scale.

```
# select top 2 models based on BIC and RMSPE
top_bic <- comparison_table[order(comparison_table$BIC),][1:2,]
top_rmspe <- comparison_table[order(comparison_table$RMSPE, decreasing = TRUE),][1:2,]
cat("\nTop 2 Models by BIC:\n")
print(top_bic[, c("Model_Name", "BIC", "RMSPE")])
cat("\nTop 2 Models by RMSPE:\n")
print(top_rmspe[, c("Model_Name", "BIC", "RMSPE")])
```

```
##
## Top 2 Models by BIC:
##              Model_Name       BIC     RMSPE
## 2 ARIMA(1,0,0)(1,0,1)12 -1811.669 0.6223465
## 8 ARIMA(1,0,1)(1,0,1)12 -1811.186 0.5961513
##
## Top 2 Models by RMSPE:
##               Model_Name       BIC     RMSPE
## 17 ARIMA(4,0,2)(1,0,1)12 -1795.560 0.6263256
## 19 ARIMA(0,0,3)(1,0,1)12 -1799.133 0.6251894
```

4 Model Shortlist

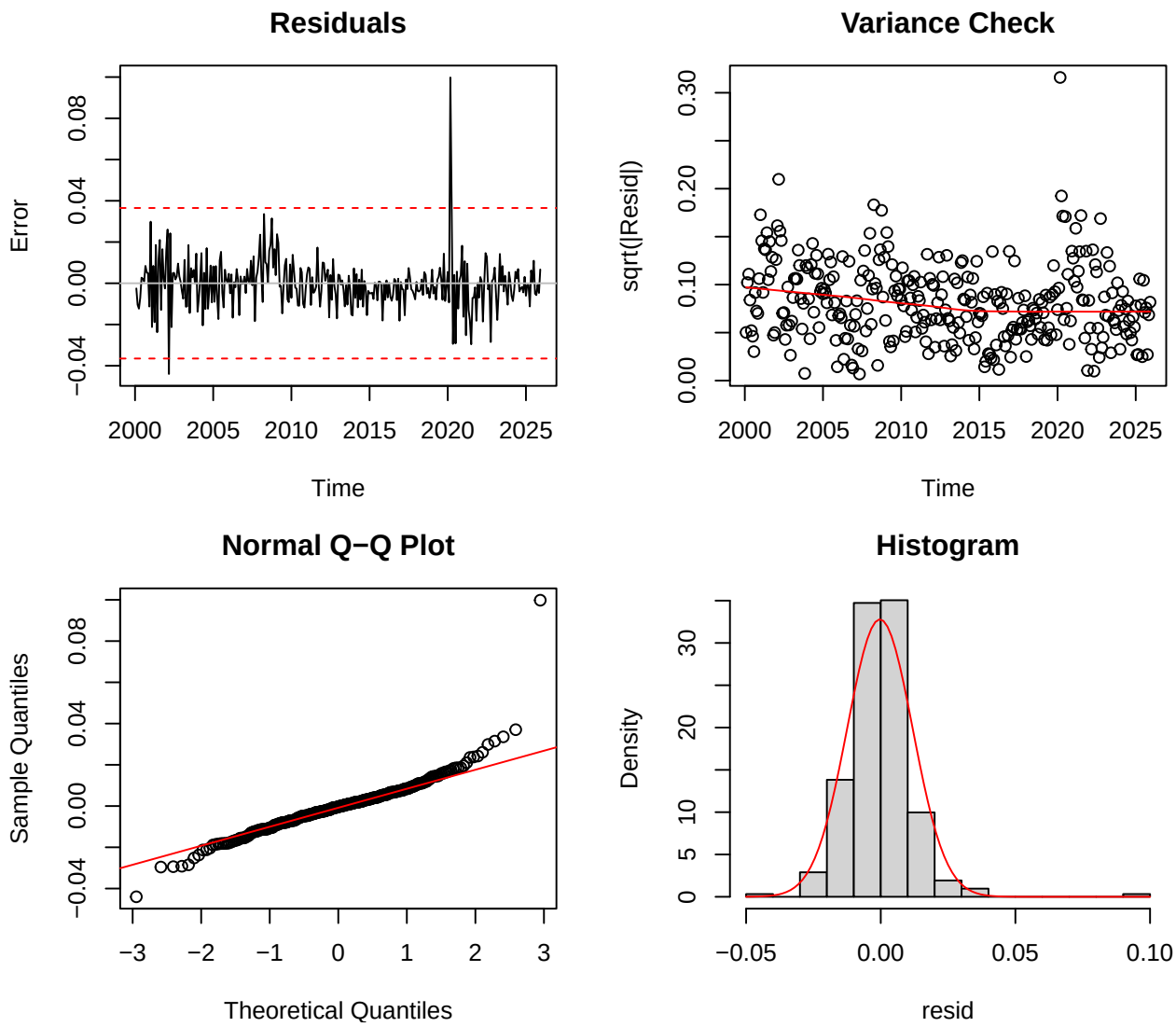
Now we analyze the shortlisted models in more detail. ## Function

```
analyze_sarima_wrapper <- function(model) {
  chosen_model_index <- which(comparison_table$Model_Name == model)
  # geet specs from my_models based on index
  chosen_model_spec <- my_models[[chosen_model_index]]
  stats <- analyze_sarima(
    d1ln_transformed,
    order = chosen_model_spec$order,
    seasonal = chosen_model_spec$seasonal,
    verbose = TRUE
)
  return(stats)
}
```

```
stats <- analyze_sarima_wrapper(top_bic$Model_Name[1])
print(t(stats))
```

```
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(1,0,0)(1,0,1)12
## ==================================================================
##
```

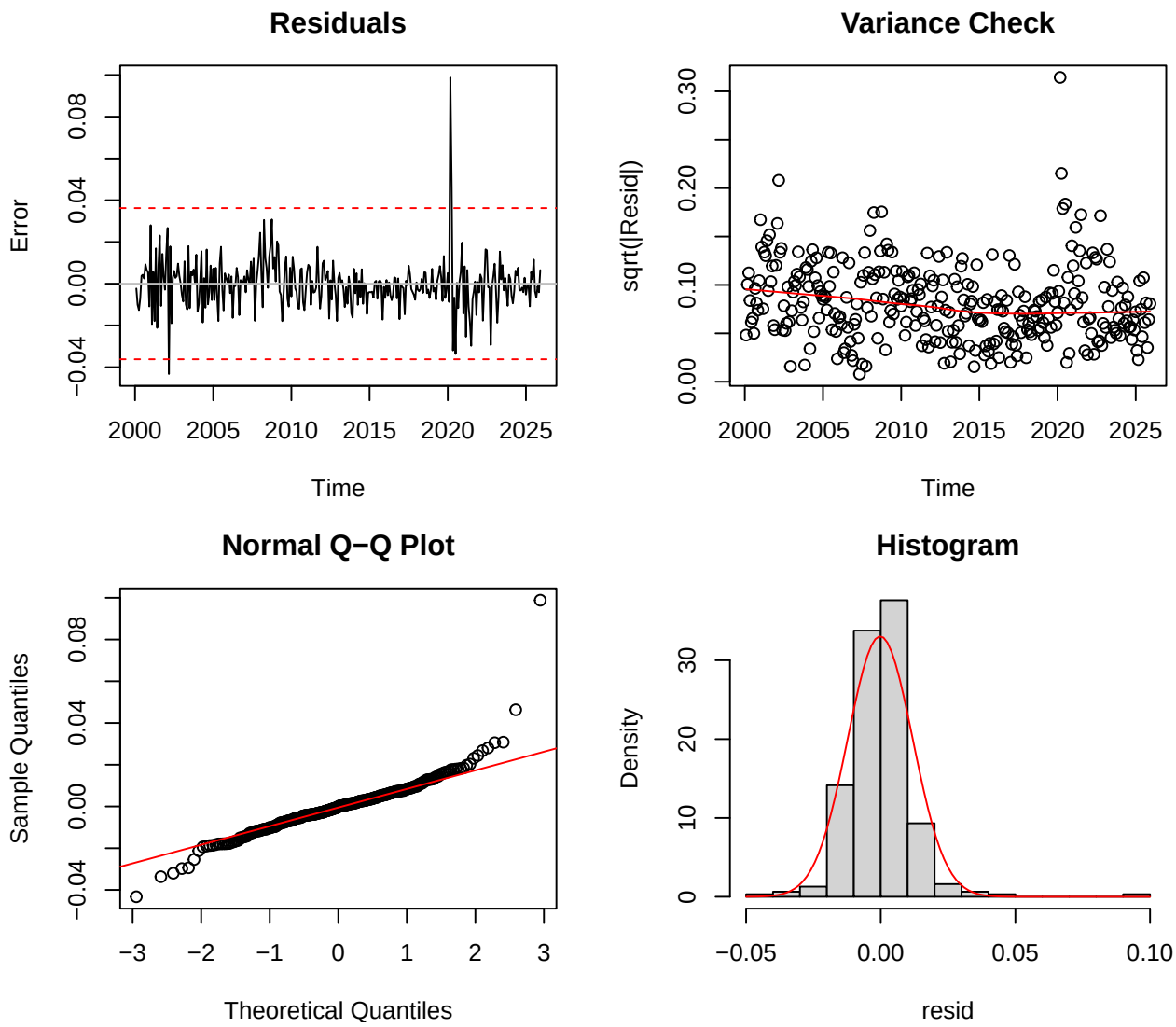
```
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.6151    0.0452  13.6191  0.0000
## sar1        0.9923    0.0062 161.3155  0.0000
## sma1       -0.8653    0.0475 -18.2217  0.0000
## intercept  -0.0003    0.0097  -0.0353  0.9719
##
## Significance Flags:
##           Significant
## ar1               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.00064
## Min MA Root Modulus:  1.012133
##                 [,1]
## Model_Name      "ARIMA(1,0,0)(1,0,1)12"
## AIC             "-1830.368"
## BIC             "-1811.669"
## Sigma2          "0.0001476447"
## Resid_Norm_PVal "3.367757e-14"
## Normal          "No"
## Resid_Ind_PVal  "0.09928266"
## Independent     "Yes"
## Min_AR_Root     "1.00064"
## Causal          "Yes"
## Min_MA_Root     "1.012133"
## Invertible      "Yes"
## RMSE            "0.00616029"
## RMSPE           "0.6223465"
## MAE             "0.005002779"
## MAPE_Pct        "0.5034827"
## Mean_CI_Width   "0.05957034"
```




```r
stats <- analyze_sarima_wrapper(top_bic$Model_Name[2])
print(t(stats))
```

```
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(1,0,1)(1,0,1)12
## ================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##            Estimate Std.Error   t_stat p_value
## ar1          0.7467    0.0633  11.7909  0.0000
## ma1         -0.2189    0.0977  -2.2402  0.0251
## sar1         0.9914    0.0065 152.6144  0.0000
## sma1        -0.8536    0.0479 -17.8350  0.0000
## intercept   -0.0006    0.0117  -0.0493  0.9607
##
## Significance Flags:
##            Significant
## ar1                Yes
```

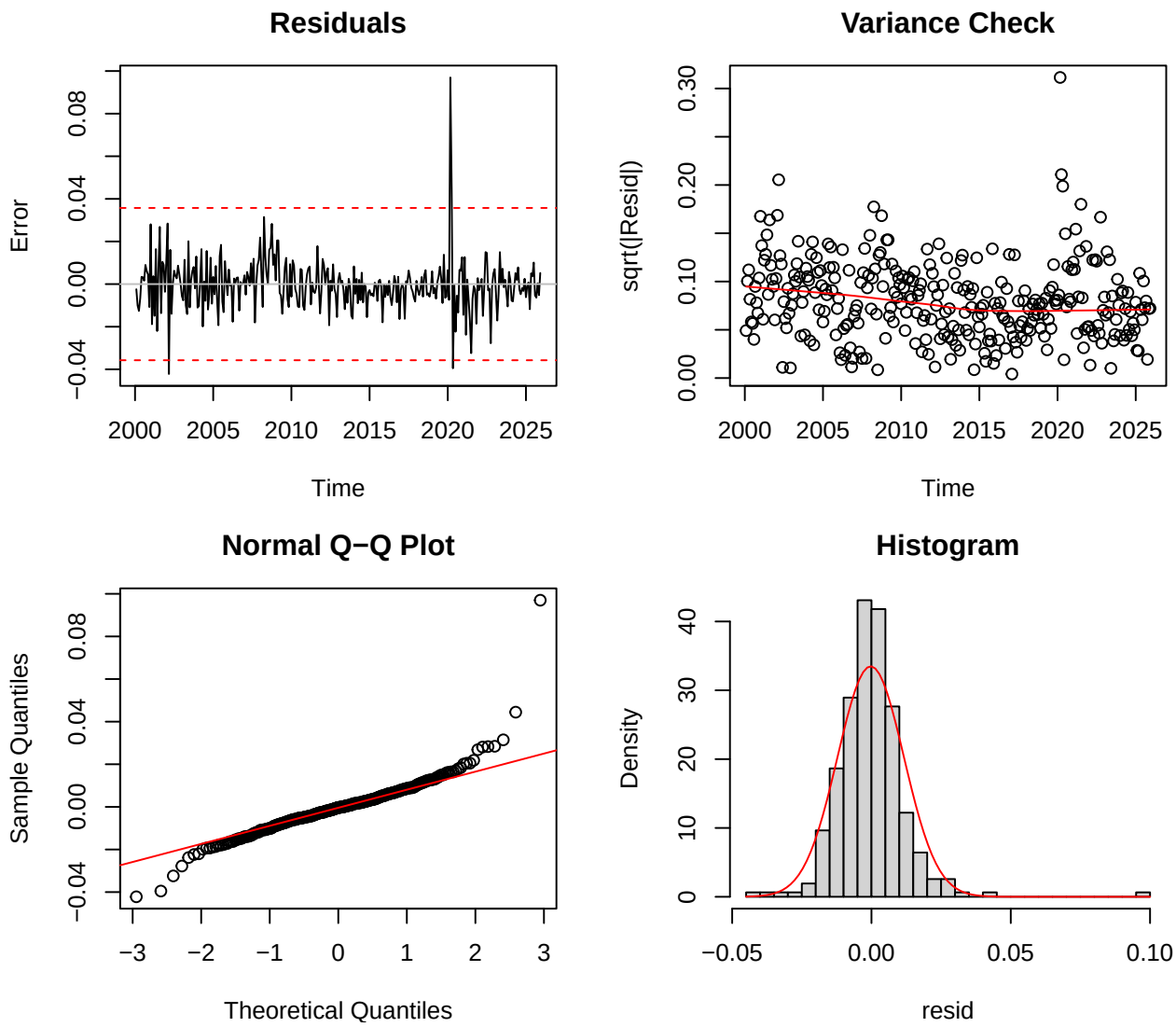
```
## ma1               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000721
## Min MA Root Modulus:  1.013275
##                 [,1]
## Model_Name      "ARIMA(1,0,1)(1,0,1)12"
## AIC             "-1833.625"
## BIC             "-1811.186"
## Sigma2          "0.0001452986"
## Resid_Norm_PVal "1.170265e-14"
## Normal          "No"
## Resid_Ind_PVal  "0.3330666"
## Independent     "Yes"
## Min_AR_Root     "1.000721"
## Causal          "Yes"
## Min_MA_Root     "1.013275"
## Invertible      "Yes"
## RMSE            "0.005897894"
## RMSPE           "0.5961513"
## MAE             "0.004715277"
## MAPE_Pct        "0.4746397"
## Mean_CI_Width   "0.05873444"
```




```r
stats <- analyze_sarima_wrapper(top_rmspe$Model_Name[1])
print(t(stats))
```

```
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(4,0,2)(1,0,1)12
## ================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##            Estimate Std.Error   t_stat p_value
## ar1          0.2166    0.1366   1.5851  0.1130
## ar2         -0.5172    0.1080  -4.7913  0.0000
## ar3          0.4244    0.1204   3.5242  0.0004
## ar4          0.1212    0.0637   1.9035  0.0570
## ma1          0.3334    0.1228   2.7154  0.0066
## ma2          0.8806    0.1154   7.6282  0.0000
## sar1         0.9926    0.0059 169.0281  0.0000
## sma1        -0.8672    0.0461 -18.8139  0.0000
## intercept   -0.0004    0.0108  -0.0362  0.9711
```

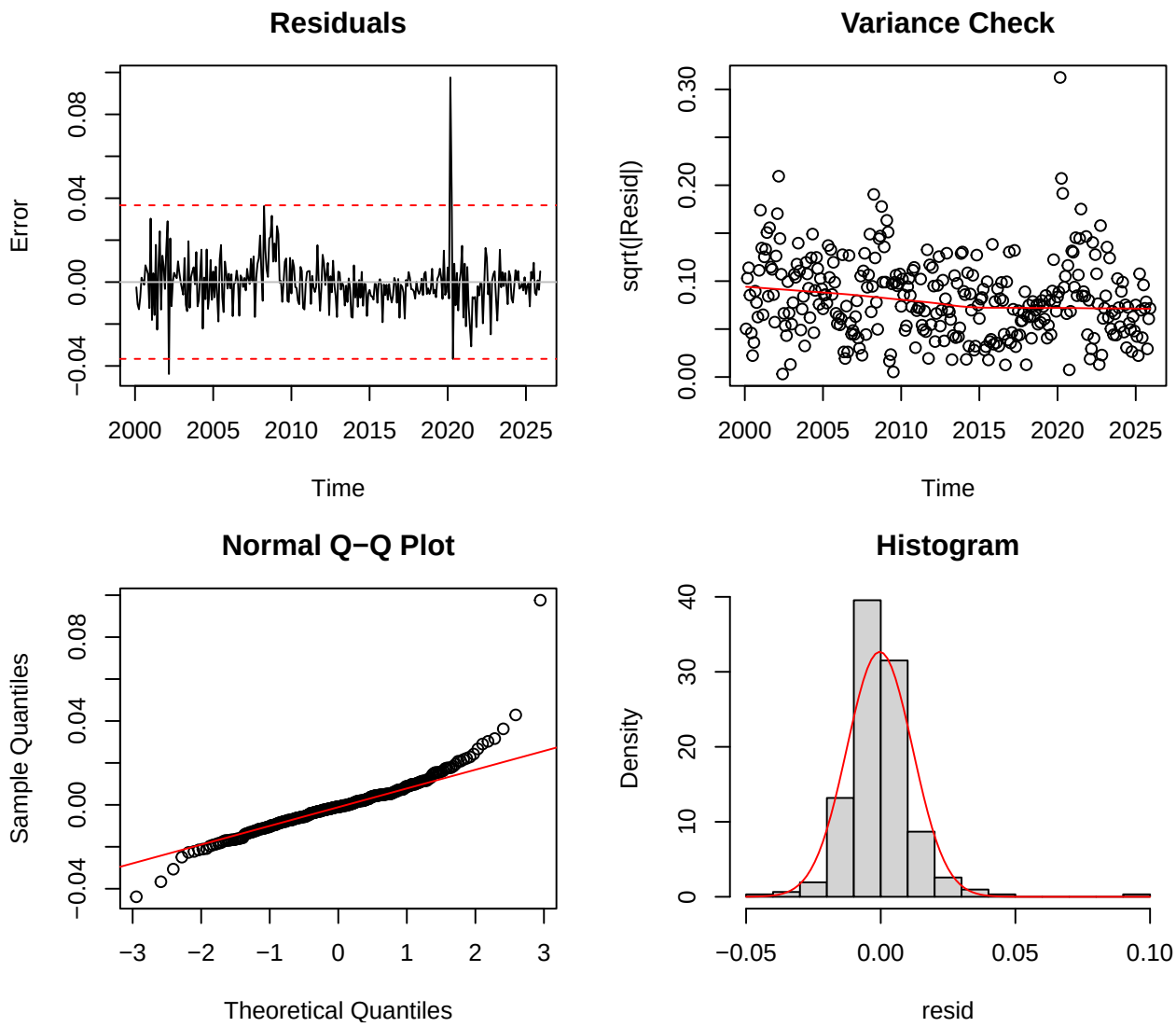
```
##
## Significance Flags:
##           Significant
## ar1                No
## ar2               Yes
## ar3               Yes
## ar4                No
## ma1               Yes
## ma2               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000619
## Min MA Root Modulus:  1.011946
##                  [,1]
## Model_Name      "ARIMA(4,0,2)(1,0,1)12"
## AIC             "-1832.958"
## BIC             "-1795.56"
## Sigma2          "0.0001415366"
## Resid_Norm_PVal "1.284736e-14"
## Normal          "No"
## Resid_Ind_PVal  "0.6870367"
## Independent     "Yes"
## Min_AR_Root     "1.000619"
## Causal          "Yes"
## Min_MA_Root     "1.011946"
## Invertible      "Yes"
## RMSE            "0.006196291"
## RMSPE           "0.6263256"
## MAE             "0.004882362"
## MAPE_Pct        "0.4911521"
## Mean_CI_Width   "0.05867902"
```




```r
stats <- analyze_sarima_wrapper(top_rmspe$Model_Name[2])
print(t(stats))
```

```
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(0,0,3)(1,0,1)12
## ================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##            Estimate Std.Error   t_stat p_value
## ma1          0.5659    0.0573   9.8768  0.0000
## ma2          0.4406    0.0615   7.1602  0.0000
## ma3          0.1560    0.0486   3.2093  0.0013
## sar1         0.9911    0.0072 137.5903  0.0000
## sma1        -0.8610    0.0502 -17.1677  0.0000
## intercept   -0.0006    0.0078  -0.0713  0.9431
##
## Significance Flags:
##            Significant
```

```
## ma1                Yes
## ma2                Yes
## ma3                Yes
## sar1               Yes
## sma1               Yes
## intercept           No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000742
## Min MA Root Modulus:  1.012547
##                  [,1]
## Model_Name       "ARIMA(0,0,3)(1,0,1)12"
## AIC              "-1825.312"
## BIC              "-1799.133"
## Sigma2           "0.0001487181"
## Resid_Norm_PVal "2.595307e-14"
## Normal           "No"
## Resid_Ind_PVal   "0.01592004"
## Independent      "No"
## Min_AR_Root      "1.000742"
## Causal           "Yes"
## Min_MA_Root      "1.012547"
## Invertible       "Yes"
## RMSE             "0.006185493"
## RMSPE            "0.6251894"
## MAE              "0.00506944"
## MAPE_Pct         "0.5103412"
## Mean_CI_Width    "0.05872291"
```



4.1 Final Model: ARIMA(1,0,0)(1,0,1)[12]

Across all models, the residuals fail the normality test, which is also evident in the Q–Q plots. However, the residual plots and histograms suggest that this deviation is largely driven by extreme observations during the Covid period. The 2008 financial crisis shows elevated residuals as well, but not to the same magnitude. We also observe one additional extreme residual between 2000 and 2005 that cannot be linked to any identifiable event. Regarding variance, although it decreases slightly over time, the slope is minimal, indicating that the variance is relatively stable overall. The variance-diagnostic plot does show signs of heteroskedasticity, likely reflecting structural breaks in the data. In terms of coefficient significance, the ARIMA(1,0,0)(1,0,1)[12] and ARIMA(0,0,3)(1,0,1)[12] models have all coefficients statistically significant, whereas the remaining models include non-significant terms. Selecting the most parsimonious model with fully significant coefficients, we choose ARIMA(1,0,0)(1,0,1)[12] as the final specification.